Customer Feedback Analysis using Natural Language Processing

Nov 4, 2026 · @December

Abstract

A fine-tuned DistilBERT model classified Amazon customer reviews into negative, neutral and positive sentiment with a macro F1 score of 0.717, the best of six models compared. Businesses receive far more customer feedback than people can read, so this project builds an end-to-end Natural Language Processing (NLP) pipeline that reads it automatically, using only free and open-source models.

The dataset contains 20,913 cleaned public reviews of Amazon written between 2007 and 2024. The pipeline cleans the text, explores the data, and compares a rule-based model (VADER), three classical machine learning models (TF-IDF with Naive Bayes, Logistic Regression and Linear SVM), a pretrained transformer (RoBERTa) and a fine-tuned transformer (DistilBERT). It then extracts insights with topic modeling (BERTopic), zero-shot categorization (BART-MNLI), aspect-based sentiment analysis (DeBERTa) and summarization (BART-CNN). A Streamlit dashboard makes the system usable by non-programmers.

Fine-tuning improved macro F1 by 4.4 points over the best classical model and by 7.2 points over the pretrained transformer, and nearly doubled the F1 score on the hardest, neutral class (0.285 against 0.156). In the dashboard, the model's predictions agreed with customers' star ratings for 93% of a random sample of 300 reviews. The analysis also shows a sharp decline in customer satisfaction: the average rating fell from about 4.0 stars in 2016–2017 to about 1.6 stars in 2021–2024, with delivery, refunds and customer service dominating the complaints.

Keywords: sentiment analysis, customer feedback, transformers, DistilBERT, topic modeling, aspect-based sentiment analysis

1. Introduction

1.1 Background

Customer feedback is one of the most direct signals a business gets about what is working and what is not. Online reviews, survey comments and support messages arrive continuously and in large volumes. A single company can collect thousands of reviews a month, far more than a team can read and summarize by hand.

Natural Language Processing (NLP) lets a computer read this text at scale. It can decide whether a review is positive or negative, group reviews by subject, and summarize recurring complaints. Recent transformer models such as BERT have made these tasks much more accurate, and many of them are now freely available.

1.2 Problem statement

Given a large collection of raw, unstructured customer reviews, automatically determine:

1. The overall sentiment of each review (negative, neutral or positive).
2. The main themes customers write about.
3. The sentiment toward specific aspects of the service, such as delivery or refunds, within a single review.
4. A short, readable summary of the most common complaints.

1.3 Objectives

- Clean and explore a real-world customer review dataset.
- Build and compare sentiment classifiers across three levels: rule-based, classical machine learning and transformers.
- Fine-tune a transformer on the dataset and measure the gain over off-the-shelf models.
- Discover topics and business categories, and analyze sentiment at the aspect level.
- Generate summaries that turn complaints into actionable insights.
- Deliver the system as an interactive web dashboard.

1.4 Scope and constraints

The project works with English-language reviews of Amazon as a company. All models and libraries are free and open source, and all training runs on the free tier of Google Colab with a single T4 GPU. Sentiment is derived from the customers' own star ratings, so no manual labeling was needed.

2. Literature Review

Sentiment analysis has moved from hand-built word lists, to statistical models over word counts, to pretrained transformers that read words in context. This project compares one representative of each stage on the same data.

2.1 Lexicon-based methods

Lexicon methods score text using a dictionary of words with known sentiment. VADER (Hutto and Gilbert, 2014) adds rules for negation, intensifiers, capital letters and punctuation, and was designed for short social media text. It needs no training data, but it cannot learn domain-specific meaning, and it often misreads long reviews that mix polite words with complaints.

2.2 Classical machine learning

Classical approaches turn each document into a vector of word weights, most commonly with TF-IDF (term frequency–inverse document frequency), and train a classifier on labeled examples. Pang, Lee and Vaithyanathan (2002) showed that Naive Bayes and Support Vector Machines outperform lexicon baselines on movie reviews. Word pairs (bigrams) let these models capture short phrases such as "not good". They are fast and interpretable, but they ignore word order beyond a few words.

2.3 Transformer models

The Transformer architecture (Vaswani et al., 2017) uses self-attention to model the relationship between every pair of words in a text. BERT (Devlin et al., 2019) pretrains a transformer on large text collections and is then fine-tuned for a specific task with a small labeled dataset. RoBERTa (Liu et al., 2019) improves BERT's pretraining. DistilBERT (Sanh et al., 2019) is 40% smaller and 60% faster than BERT while keeping about 97% of its language understanding, which makes it practical on free hardware. The Twitter-RoBERTa sentiment model (Barbieri et al., 2020; Loureiro et al., 2022) is a RoBERTa model already trained for three-class sentiment on tweets.

2.4 Topic modeling

Latent Dirichlet Allocation (LDA; Blei, Ng and Jordan, 2003) finds topics from word co-occurrence counts. BERTopic (Grootendorst, 2022) instead clusters sentence embeddings, so reviews with similar meanings group together even when they share no words, and then describes each cluster with class-based TF-IDF keywords.

2.5 Zero-shot classification and aspect-based sentiment

Yin, Hay and Roth (2019) showed that a model trained on natural language inference can classify text into labels it has never seen, by testing whether the text implies a statement such as "This review is about delivery". Aspect-based sentiment analysis (ABSA; Pontiki et al., 2014) goes further than one label per review and finds the sentiment toward each aspect mentioned, which matters for mixed reviews.

2.6 Summarization

BART (Lewis et al., 2020) is a sequence-to-sequence transformer that performs well on abstractive summarization, where the model writes new sentences rather than copying them. The BART-large-CNN checkpoint is fine-tuned on news articles.

2.7 Research gap

Many student projects stop at overall sentiment on a balanced benchmark dataset. This project uses a real, heavily imbalanced dataset, evaluates with macro F1 rather than accuracy, and connects sentiment results to topics, aspects and summaries that a business could act on.

3. Dataset

The project uses 20,913 cleaned customer reviews of Amazon, 69% of them negative, taken from the file Amazon_Reviews.csv.

3.1 Source and structure

The file contains public reviews of Amazon as a company (not of individual products), written between August 2007 and September 2024. Each record has nine columns:

Column	Description	Used?
Reviewer Name	Display name of the reviewer	No (personal data)
Profile Link	Link to the reviewer's profile	No (personal data)
Country	Two-letter country code	Yes, for EDA
Review Count	Number of reviews the person has written	No
Review Date	Date and time of the review	Yes, for trends
Rating	Text such as "Rated 4 out of 5 stars"	Yes, as the label
Review Title	Short headline	Yes, joined to the text
Review Text	Full review	Yes
Date of Experience	Date of the experience described	No

3.2 Data cleaning

Step	Problem found	Action	Records
Parse the file	267 lines were fragments of reviews broken across lines	Kept only rows with all 9 columns	20,947 complete reviews
Extract rating	Rating stored as text	Took the digit from "Rated N out of 5 stars"	All rows
Build review text	48% of titles were truncated copies of the text's first line	Joined title and text, skipping a duplicate title	All rows
Remove duplicates	Repeated review texts	Dropped duplicates	20,913 reviews
Remove personal data	Names and profile links	Dropped those columns	—
Placeholder text (found in Phase 8)	630 reviews had only a title and the text "Review text not found"	Removed the placeholder before topic modeling	630 affected

3.3 Labels

Sentiment labels come from the customers' own star ratings:

Stars	Sentiment	Reviews	Share
1–2	Negative	14,338	68.6%
3	Neutral	871	4.2%
4–5	Positive	5,704	27.3%

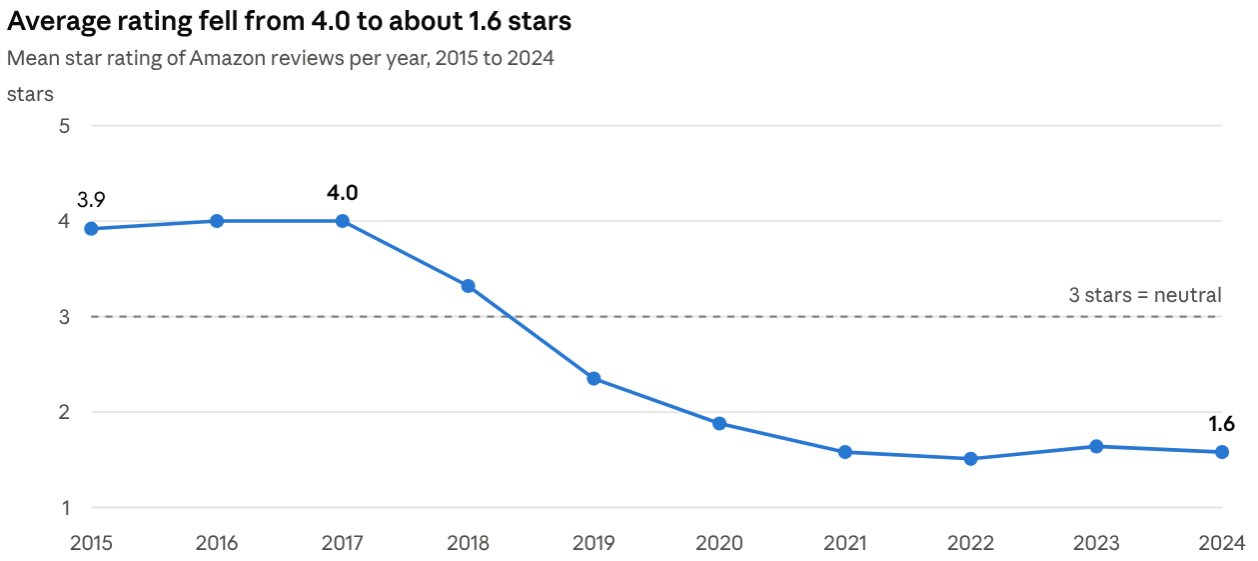
Three classes were used instead of five because neighbouring ratings (1 and 2 stars, or 4 and 5 stars) are hard to tell apart from text, and three classes match how a business acts on feedback.

3.4 Class imbalance

The neutral class is only 4.2% of the data. A model that always predicts "negative" would already reach 68.6% accuracy while being useless. This shaped two design decisions: the training set is rebalanced (Section 5.3), and models are compared by macro F1, which weights all three classes equally (Section 5.6).

4. Exploratory Data Analysis

The most important finding of the EDA is a collapse in customer satisfaction: the average rating fell from 4.0 stars in 2016–2017 to about 1.6 stars from 2021 onwards.



Amazon_Reviews.csv · mean stars per year after cleaning, 2015–2024

Ratings dropped steeply between 2018 and 2020 and have stayed below the neutral line since. Reviews before 2015 are too few per year to compare reliably.

4.1 Rating distribution

Of the 20,947 complete reviews, 13,120 give 1 star and 4,448 give 5 stars, while 2, 3 and 4 stars together account for only 3,379. Customers mostly write when they are very unhappy or very happy.

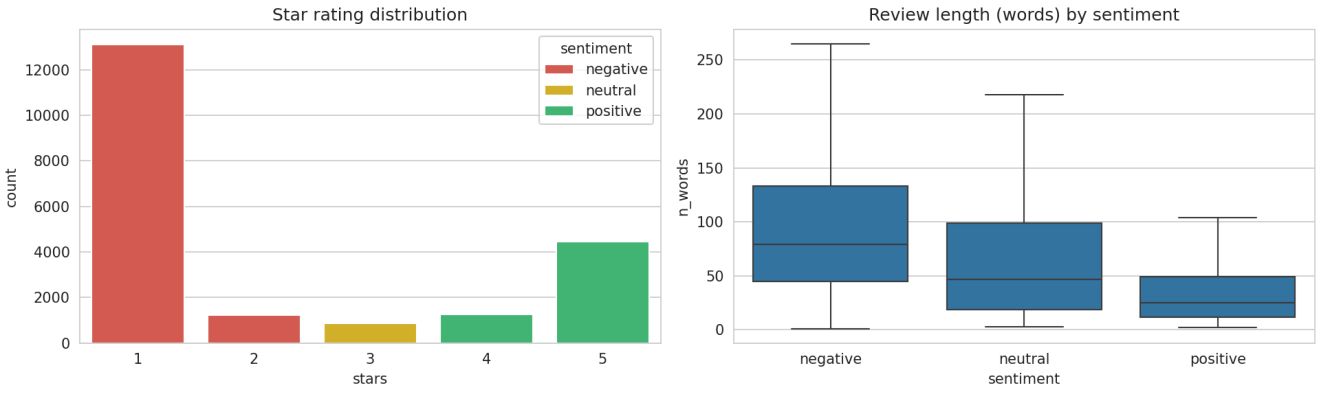


Figure 4.1: Star rating distribution (left) and review length in words by sentiment (right).

4.2 Review length

Unhappy customers write much more. The median review is 79 words for negative reviews, 47 for neutral and 25 for positive ones. Long complaints usually describe a sequence of events, such as a late delivery followed by a failed refund.

4.3 Reviews over time and by country

Most reviews were written between 2019 and 2024, peaking at about 4,000 in 2023. Reviewers are mainly from the United States (9,260) and the United Kingdom (7,238), followed by Canada (706) and India (628).

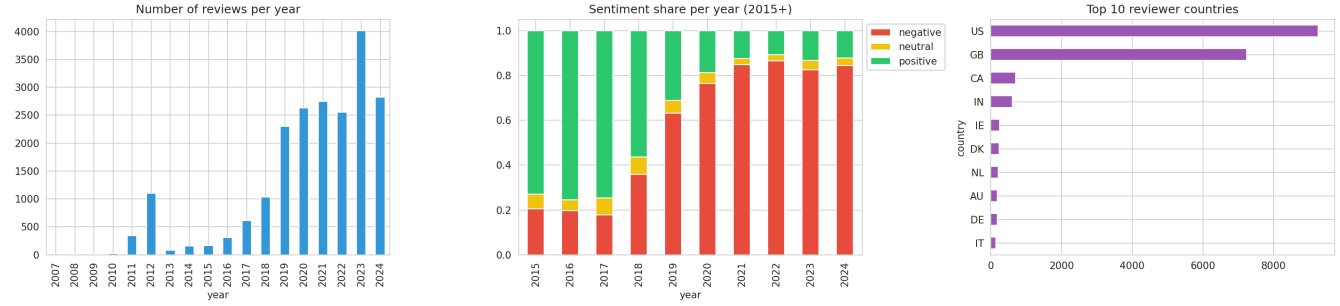


Figure 4.2: Reviews per year (left), sentiment share per year from 2015 (centre), and the top 10 reviewer countries (right).

The centre panel shows the same decline as the rating chart above, from the class point of view: negative reviews were about 20% of each year's reviews in 2015–2017, then rose to 63% in 2019 and to roughly 85% from 2021 onwards.

4.4 Word clouds

In negative reviews, the largest words are customer service, item, account, order, delivered, refund and package, which describe service failures and money. Neutral reviews are dominated by neutral shopping words such as item, order, product and time. Positive reviews feature always, love, best, easy and good.

One observation matters for modeling: customer service, order and prime appear prominently in all three clouds. The same words are used in praise and in complaints, so a model must read the words around them ("customer service was excellent" versus "customer service was useless"). This is a reason context-aware transformers outperform word-count models (Section 6).

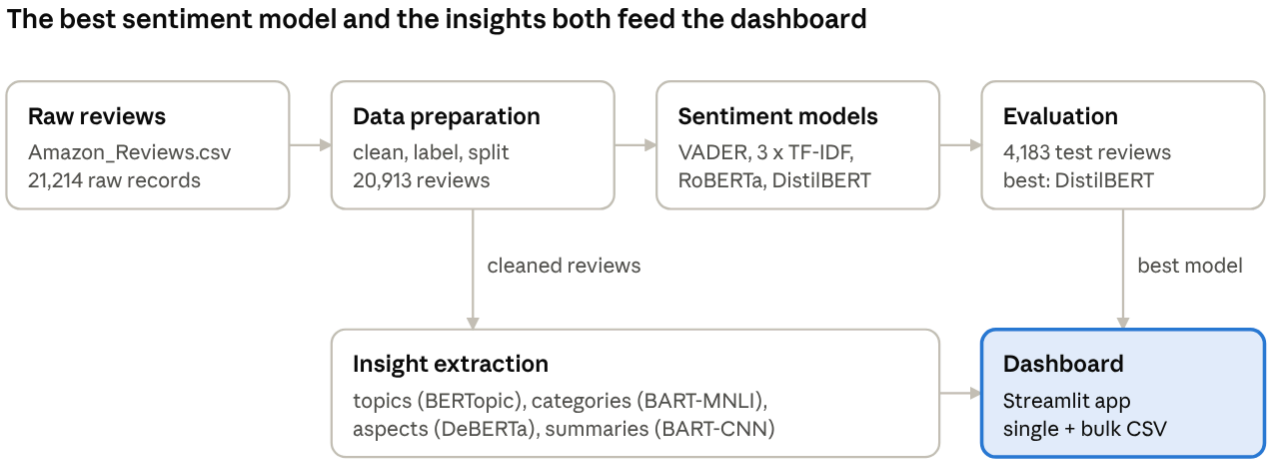


Figure 4.3: Word clouds for negative, neutral and positive reviews (common English stopwords and the word "amazon" removed).

5. Methodology

The system is a 13-phase pipeline grouped into five stages: data preparation, sentiment modeling, evaluation, insight extraction and deployment.

5.1 System architecture



system architecture · 6 components, Phases 0–12

Cleaned reviews feed both the sentiment models and the insight extraction; the best model from the evaluation and the insights come together in the dashboard.

Stage	Phases	Purpose
1. Data preparation	0–3	Setup, loading, EDA, preprocessing
2. Sentiment modeling	4–6	Baselines, pretrained and fine-tuned transformers
3. Evaluation	7	Model comparison and error analysis

Stage	Phases	Purpose
4. Insight extraction	8–10	Topics, categories, aspects, summaries
5. Deployment	11–12	Model export and Streamlit dashboard

5.2 Text preprocessing

Classical models need clean, normalized text, so each review passes through these steps:

1. Convert to lowercase.
2. Remove URLs and HTML tags.
3. Expand negations ("didn't" becomes "did not").
4. Remove numbers, punctuation and emojis.
5. Remove stopwords, but keep negation words (not, no, never, nor), since removing them would turn "not good" into "good". The word "amazon" is also removed because it appears in almost every review.
6. Lemmatize each word with WordNet ("deliveries" becomes "delivery").

For example, "I didn't like it at all" becomes "did not like". Transformer models receive the raw text instead, because they use punctuation, casing and word order as context.

5.3 Data split and rebalancing

The data is split with stratified sampling, so every split keeps the original class proportions. Only the training set is then rebalanced: the negative class is down-sampled to 5,000 reviews and the neutral class is over-sampled (with replacement) to 2,500. The validation and test sets keep the real distribution, so results reflect real-world performance.

Split	Share	Negative	Neutral	Positive	Total
Train (before rebalancing)	72%	10,323	627	4,107	15,057
Train (after rebalancing)	—	5,000	2,500	4,107	11,607
Validation	8%	—	—	—	1,673
Test	20%	—	—	—	4,183

5.4 Sentiment models

Model	Type	Input	Training on this data
VADER	Rule-based lexicon	Raw text	None
TF-IDF + Naive Bayes	Classical ML	Cleaned text	Yes
TF-IDF + Logistic Regression	Classical ML	Cleaned text	Yes
TF-IDF + Linear SVM	Classical ML	Cleaned text	Yes
RoBERTa (cardiffnlp/twitter-roberta-base-sentiment-latest)	Pretrained transformer	Raw text	None
DistilBERT (distilbert-base-uncased)	Fine-tuned transformer	Raw text	Yes

VADER labels a review positive when its compound score is at least 0.05, negative when it is at most −0.05, and neutral otherwise. The TF-IDF vectorizer keeps the 30,000 most useful single words and word pairs that appear in at least two reviews, with sublinear term frequency. It is fitted on the training set only, to avoid leaking test data into training.

5.5 Fine-tuning setup

Hyperparameter	Value	Reason
Base model	distilbert-base-uncased (66M parameters)	Fast enough for a free GPU
Epochs	2	Enough to adapt without overfitting
Learning rate	2×10^{-5}	Small, so pretrained knowledge is kept
Batch size	16 (train), 64 (evaluation)	Fits T4 GPU memory
Max sequence length	256 tokens	Covers most reviews
Warmup	First 10% of training steps	Stable start

Hyperparameter	Value	Reason
Weight decay	0.01	Regularization
Precision	Mixed (fp16)	Faster training on GPU
Model selection	Best validation macro F1 across epochs	Guards against overfitting

5.6 Evaluation metrics

All models are evaluated on the same held-out test set of 4,183 reviews. Because the classes are imbalanced, the main metric is macro F1: the F1 score is computed for each class separately and then averaged, so the small neutral class counts as much as the large negative class.

```
\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \qquad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \qquad F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}
```

```
\text{Macro } F1 = \frac{1}{3} \sum_{c \in \{\text{neg},\, \text{neu},\, \text{pos}\}} F1_c
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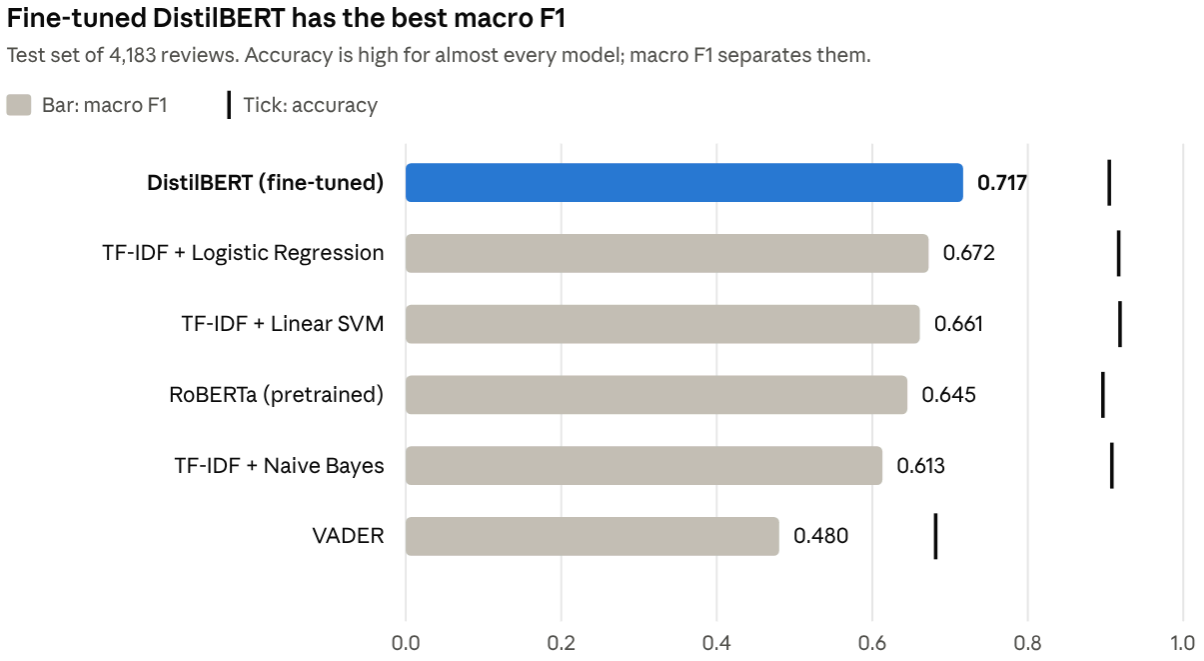
Accuracy, macro precision and macro recall are also reported, together with a confusion matrix for each model.

5.7 Insight extraction methods

- **Topic modeling:** BERTopic with all-MiniLM-L6-v2 sentence embeddings, UMAP dimensionality reduction and K-Means clustering into 12 topics, on a sample of 6,000 reviews of at least 8 words. Section 8.1 explains why K-Means replaced the default HDBSCAN clustering.
- **Categorization:** zero-shot classification with facebook/bart-large-mnli into seven categories: delivery, customer service, refund and returns, account and payment, Prime membership, product quality and price. A review gets every category scoring above 0.6, or its single best category, on a sample of 400 test reviews.
- **Aspect-based sentiment:** yangheng/deberta-v3-base-absa-v1.1, which takes a review and an aspect and returns the sentiment toward that aspect.
- **Summarization:** facebook/bart-large-cnn summarizes up to 12 detailed negative reviews (20 words or more) from each of the five most-complained-about topics.

6. Results

The fine-tuned DistilBERT achieved the highest macro F1 of 0.717, ahead of TF-IDF + Logistic Regression (0.672) and the pretrained RoBERTa (0.645).



outputs/model_comparison.csv · 6 models on the 4,183-review test set

6.1 Full comparison

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
DistilBERT (fine-tuned)	0.905	0.704	**0.739**	**0.717**
TF-IDF + Logistic Regression	0.917	0.694	0.666	0.672
TF-IDF + Linear SVM	**0.919**	0.694	0.658	0.661
RoBERTa (pretrained)	0.897	0.652	0.639	0.645
TF-IDF + Naive Bayes	0.908	**0.715**	0.618	0.613
VADER	0.682	0.493	0.527	0.480

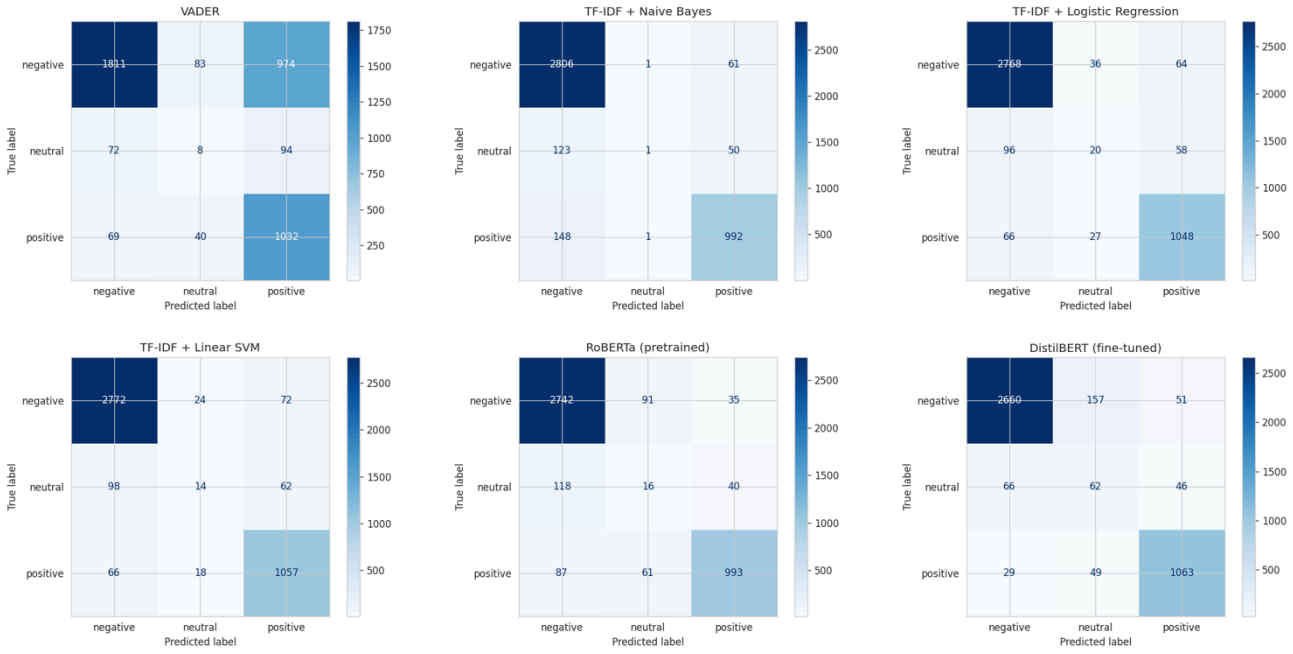


Figure 6.1: Confusion matrices of all six models on the 4,183-review test set. Rows are true labels, columns are predicted labels.

6.2 Per-class performance

The neutral class separates the models: fine-tuned DistilBERT reaches a neutral F1 of 0.285, almost twice the best classical model (0.156).

Model	F1 negative	F1 neutral	F1 positive	Neutral reviews found (of 174)
DistilBERT (fine-tuned)	0.946	**0.285**	**0.923**	**62 (35.6%)**
TF-IDF + Logistic Regression	**0.955**	0.156	0.907	20 (11.5%)
TF-IDF + Linear SVM	**0.955**	0.122	0.907	14 (8.0%)
RoBERTa (pretrained)	0.943	0.094	0.899	16 (9.2%)
TF-IDF + Naive Bayes	0.944	0.011	0.884	1 (0.6%)
VADER	0.751	0.052	0.637	8 (4.6%)

All trained models score above 0.94 on negative reviews and around 0.90 on positive reviews, so the differences in macro F1 come almost entirely from the neutral class. Naive Bayes recognized only 1 of the 174 neutral test reviews, because with very few neutral examples its word probabilities are dominated by the other two classes. DistilBERT found

62, more than three times as many as any other model. It does this at a small cost: it labels 157 negative reviews as neutral, which lowers its negative F1 slightly (0.946 against 0.955) and its overall accuracy.

6.3 Training behaviour

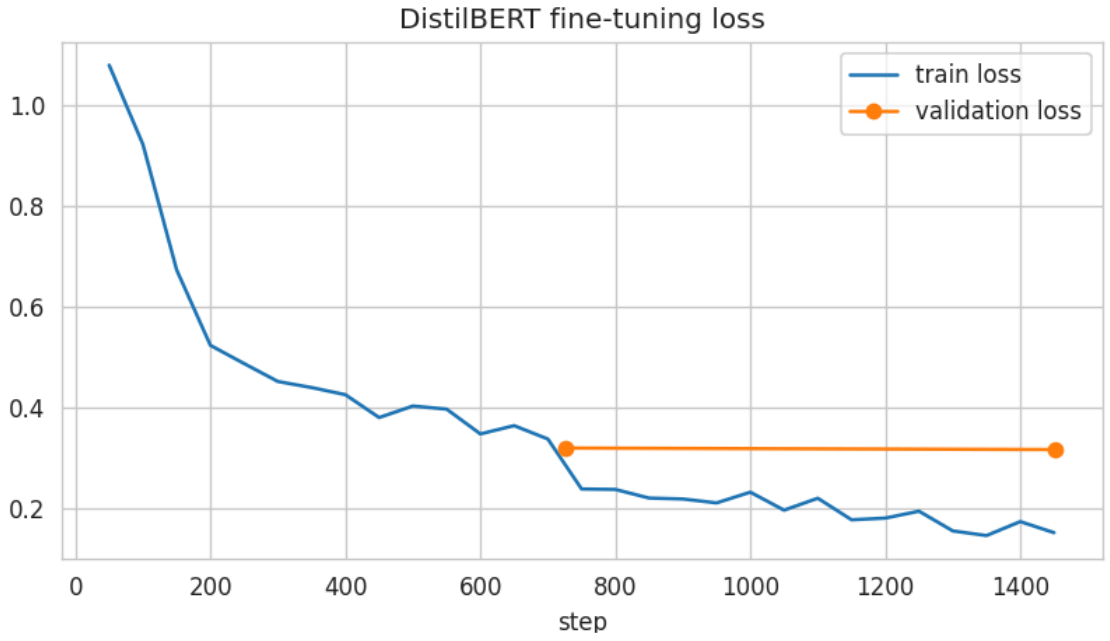


Figure 6.2: DistilBERT training loss (every 50 steps) and validation loss (after each epoch).

Training ran for 2 epochs of about 726 steps each. Training loss fell from about 1.08 to about 0.15, while validation loss stayed at about 0.32 after both epochs. The growing gap means that, in the second epoch, the model fitted the training reviews more closely without improving on unseen ones. Two epochs was therefore enough; more epochs would most likely overfit. The best epoch was selected by validation macro F1, which guards against this.

6.4 Discussion

1. **Fine-tuning gives the largest gain.** DistilBERT improved macro F1 by 4.4 points over the best classical model and by 7.2 points over the pretrained RoBERTa, and nearly doubled the F1 on the hardest (neutral) class. Adapting a transformer to the domain's own vocabulary and label definitions matters more than starting from a larger pretrained model.
2. **Accuracy is misleading on imbalanced data.** The Linear SVM has the highest accuracy (0.919) but only the third-best macro F1. DistilBERT has slightly lower accuracy (0.905) but the highest macro recall (0.739), meaning it has the best average recall across the three classes rather than relying on the majority class. Accuracy rewards getting the majority negative class right; macro F1 rewards every class equally.

3. **Classical models are strong, cheap baselines.** TF-IDF + Logistic Regression trains in seconds on a CPU and comes within 0.044 macro F1 of the fine-tuned transformer. For a business with no GPU, it would be a reasonable first system.
4. **A model trained on another domain transfers only partly.** The pretrained RoBERTa was trained on tweets. Without seeing any Amazon reviews it still reached 0.897 accuracy, but its macro F1 (0.645) is below all three trained classical models except Naive Bayes.
5. **Rule-based scoring is not enough.** VADER is the weakest model by a wide margin (macro F1 0.480). Long complaints often contain polite or positive words ("I have always loved Amazon, but…"), which a word-list approach scores as positive. VADER labeled 974 of the 2,868 negative test reviews (34%) as positive.

7. Error Analysis

The best model misclassified 398 of the 4,183 test reviews (9.5%), and 80% of those errors involve the small neutral class.

7.1 Where the errors fall

The gap between accuracy (0.905) and macro F1 (0.717) shows that errors are concentrated in one class. The test set holds only 174 neutral reviews, so even a modest number of neutral mistakes lowers the neutral F1 sharply, while the large negative class stays accurate.

True label	Predicted label	Reviews	Share of errors
Negative	Neutral	157	39.4%
Neutral	Negative	66	16.6%
Negative	Positive	51	12.8%
Positive	Neutral	49	12.3%
Neutral	Positive	46	11.6%
Positive	Negative	29	7.3%
Total		**398**	**100%**

Table 7.1: The 398 misclassifications of fine-tuned DistilBERT, from its confusion matrix (Figure 6.1).

Three findings stand out:

- **Neutral is the boundary.** 318 of the 398 errors (80%) have neutral as either the true or the predicted label. Only 80 errors (20%) confuse the two opposite classes.
- **The model over-uses neutral.** It predicted neutral 268 times, but only 62 of those were correct (23% precision). The largest single error is 157 negative reviews labeled neutral. Many of these are likely mild or mixed 1–2 star complaints.
- **Opposite mistakes are rare.** Only 51 negative reviews were called positive, compared with 974 for VADER. Fine-tuning taught the model that polite wording inside a complaint does not make it positive.

7.2 Types of errors

Misclassified reviews typically fall into the patterns below. The examples are illustrative, written to show each pattern clearly.

Error type	Illustrative example	Why the model fails
Neutral confusion	"Delivery is okay, but prices have gone up."	Three-star reviews mix praise and complaints, so they sit between the other classes
Mixed opinions	"Love the products, but customer service is useless."	One label cannot describe two opposite opinions
Sarcasm	"Great, another parcel 'delivered' to nowhere."	Positive words carry a negative meaning
Rating–text mismatch	A 1-star rating on "Amazon is great, only the courier was bad"	The label comes from the star rating, and the customer's rating does not always match the text
Very short or title-only reviews	"Ok." (630 reviews contained only a title)	Too little text to judge
Very long reviews	Complaints of 500 words or more	The model reads only the first 256 tokens

The confusion matrix supports the first two patterns directly: neutral confusion accounts for 80% of all errors, and mixed reviews are what three-star ratings usually contain.

7.3 What the errors imply

Two of these error types are limits of the data rather than of the model. Star ratings are a noisy label, and a single label per review cannot represent mixed opinions. This motivates the aspect-based sentiment analysis in Section 8.3, which scores each aspect of a review separately.

8. Insight Extraction

Beyond positive or negative, the system finds what customers complain about: customer service, delivery and refunds are the recurring themes in this dataset.

8.1 Topic modeling

First run and the problem it revealed. With BERTopic's default HDBSCAN clustering, one topic absorbed 5,812 of the 6,000 sampled reviews (keywords: service, customer, delivery, customer service, order). The three remaining topics, of 63, 45 and 36 reviews, were grouped by the phrase "Review text not found" rather than by meaning. Investigating this revealed 630 reviews in the dataset that contain only a title and that placeholder text.

The fix. Two changes were made:

1. The placeholder text was removed, and reviews shorter than 8 words were left out of topic modeling.
2. HDBSCAN was replaced by K-Means with 12 clusters. Almost every review is an Amazon complaint, so their embeddings are close together, and a density-based method sees one dense region. K-Means forces the reviews into distinct themes.

Final topics.

With these changes, BERTopic produced 12 clearly separated topics from the 6,000 sampled reviews. Nine topics describe complaints and cover 4,564 reviews (76%); three describe praise and cover 1,436 reviews (24%). This split closely matches the dataset's 27% share of positive reviews, which suggests the topics capture real differences in what customers write about. The topic names below are short labels based on each topic's keywords.

Topic	Label	Reviews	Top keywords
0	Refunds and returns	824	refund, item, return, customer service, told
1	Praise: good service, fast delivery	693	great, good, service, delivery, fast, excellent
2	Account, payment and gift cards	690	account, card, credit, email, gift, bank
3	Praise: books, Kindle, online shopping	519	books, great, best, love, kindle, shopping
4	Prime membership and delivery speed	512	prime, day, delivery, membership, shipping

Topic	Label	Reviews	Top keywords
5	Delivery drivers and missing parcels	489	delivery, driver, door, package, parcel, address
6	Customer service as a company	477	customer service, company, money, worst, help
7	Late or split deliveries	462	delivery, day, delivered, order, package
8	Reviews and third-party sellers	438	reviews, products, sellers, people
9	Order problems	393	order, ordered, item, refund, product
10	Short complaints about support	279	customer service, worst, phone, chat, english
11	Praise: Prime and free shipping	224	prime, shipping, free, love, prices

The complaint topics fall into four groups: money (refunds, accounts and payments), delivery (Prime speed, drivers, late deliveries), support (customer service, chat and phone), and trust (reviews and third-party sellers). Delivery-related topics alone (4, 5 and 7) cover 1,463 reviews, about a quarter of the sample.

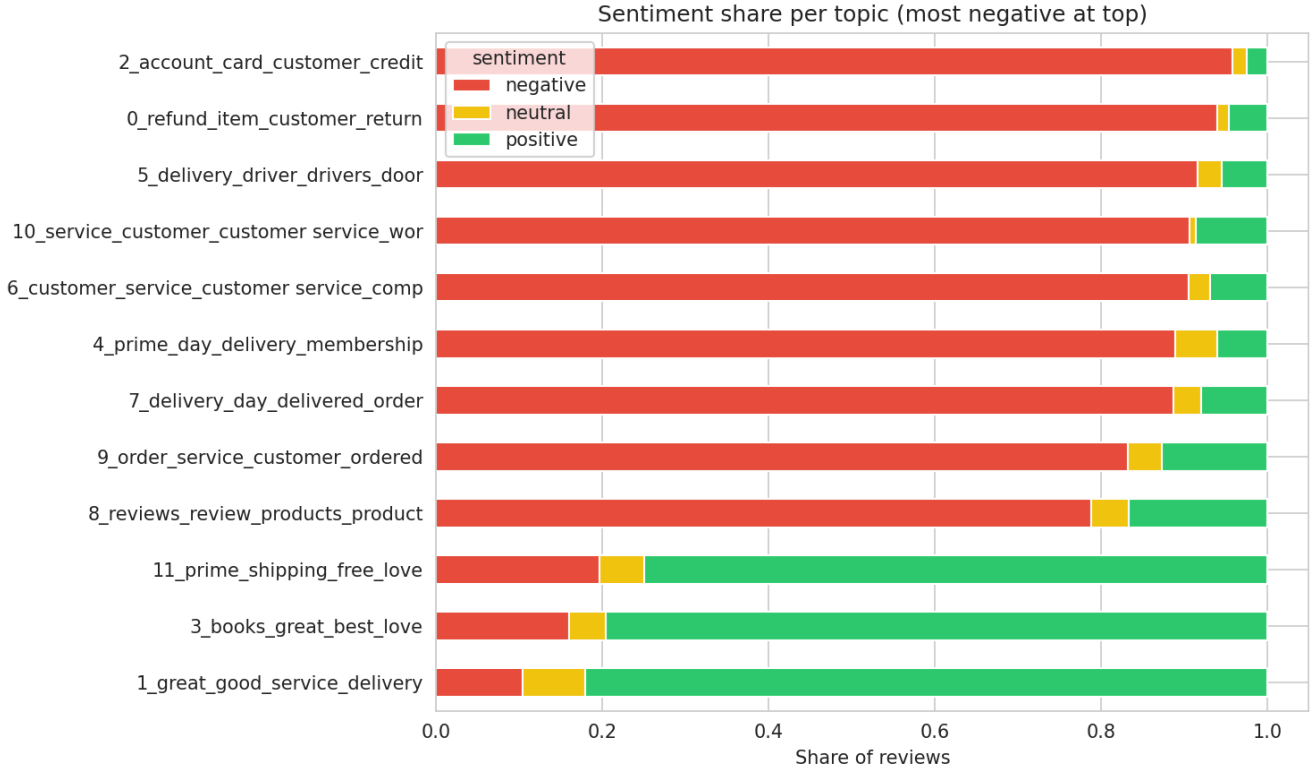


Figure 8.1: Share of negative, neutral and positive reviews in each topic, most negative at the top.

Sentiment splits sharply by topic. The account-and-payment and refund topics are the most negative, each over 90% negative, followed by delivery drivers and the two customer-service topics at about 90%. Prime, late deliveries, order problems and reviews sit between about 80% and 90% negative. The three praise topics are mostly positive, with only 10–20% negative reviews. This ranking tells a business where to look first: account security checks and refund handling generate the most uniformly negative feedback.

8.2 Feedback categorization

Zero-shot classification tagged 400 test reviews with the seven business categories without any training data. Because a review can mention several problems, each review could receive more than one category.

The same categorization also runs inside the dashboard. Figure 9.3 shows its result for 100 randomly sampled reviews, where customer service and delivery are the most-mentioned categories.

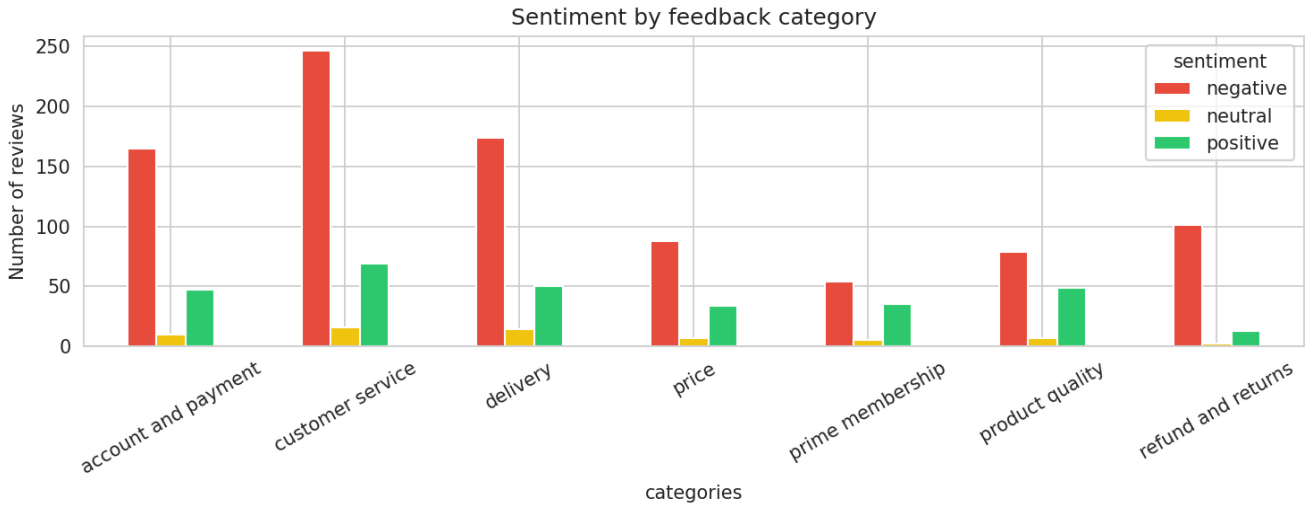


Figure 8.2: Number of negative, neutral and positive reviews per category, for 400 test reviews (a review can belong to several categories).

Customer service is the most-mentioned category, with about 245 negative reviews, followed by delivery (about 175) and account and payment (about 165). Refunds and returns is the most one-sided: about 100 negative reviews against about 12 positive. Product quality and Prime membership are the most balanced, which suggests customers are broadly satisfied with the products themselves and dissatisfied with the service around them.

Topics (Section 8.1) are discovered from the data, while categories are defined in advance by the business. Comparing the two shows whether the business's own categories match what customers actually write about.

8.3 Aspect-based sentiment

The aspect model gives each aspect of a review its own sentiment. For the review "The product quality is great but delivery was a week late and customer service kept me on hold for an hour", the dashboard returned the following:

Output	Result	Confidence
Overall sentiment (fine-tuned DistilBERT)	Neutral	0.95
Categories detected (zero-shot, score above 0.5)	product quality, delivery, customer service	—
Aspect: product quality	Positive	0.996
Aspect: delivery	Negative	0.997
Aspect: customer service	Negative	0.996

The overall label of Neutral is reasonable for a review that praises one thing and criticizes two others. On its own, though, it hides the useful information. The aspect level shows exactly what went wrong (delivery and customer service) and what went right (product quality), and each aspect is identified with very high confidence. The zero-shot model also found the right three categories and gave the other four (account and payment, price, Prime membership, refunds and returns) scores close to zero.

This directly addresses the mixed-opinion errors described in Section 7.2.

8.4 Summaries of the main complaints

The summarizer condenses detailed negative reviews from the five most-complained-about topics into a short paragraph each. A first attempt summarized short one-line reviews and copied abusive language from them, so only reviews of 20 words or more are now used.

Topic	Main complaint	Summary generated by BART-CNN
0 Refunds and returns	Product quality and recalls	"Used to be a very good company, now they have too many Eastern manufacturers selling direct on Amazon. Seems these goods are not tested. Recently had a recall on a heater we purchased via Amazon."
2 Account and payment	Accounts locked after security checks	"My Account got compromised as someone hacked my account… I wanted to know if someone could help me verify my payment information. Instead they banned me from using it."
4 Prime and delivery speed	Paying for Prime but waiting a week	"Deliver service from Amazon has fallen well below standard. That charge you extra for Amazon prime it takes over a week for the product to arrive."
5 Drivers and parcels	Contradictory information; parcels left on doorsteps	"I've made so many calls over the past 2 days and keep being told different things… Ask them to put parcels in designated safe place, not dump on doorstep."
6 Customer service	Hard to get help beyond simple returns; reviews removed	"Unless it is a straight return don't even try. When you leave a review they remove it… now they have abandoned the US customer service."

The summaries are shortened here with … and keep the customers' original wording and spelling.

Evaluation of the summaries. Four of the five summaries match their topic well and point to specific, fixable problems: account lockouts, slow Prime delivery, parcels left unsafely, and support that cannot resolve anything beyond a simple return. The summary for Topic 0 does not match its topic: it describes product quality and recalls rather than refunds. This shows why generated summaries should be checked against the topic keywords before they are shared.

BART-CNN was trained on news articles, so it often copies whole sentences from the reviews instead of rewriting them. The summaries should be read as a guide to typical complaints, not as exact statistics.

8.5 Implications for the business

- **Satisfaction has collapsed since 2018.** The average rating fell from 4.0 to about 1.6 stars, so the problems below are long-running, not isolated.
- **Service failures matter more than products.** The dominant themes are customer service, delivery, refunds and account problems, not the quality of the items sold.
- **Unhappy customers write in detail.** Negative reviews are about three times as long as positive ones, so they are a rich source of specific, fixable problems.
- **Neutral customers are rarely heard.** Only 4.2% of reviews are three-star, so a feedback survey that reaches moderate customers would give a more balanced picture.

9. Dashboard

A Streamlit web app puts the trained model in front of non-technical users: it analyzes one review instantly or a whole CSV file of reviews in bulk.

9.1 Design

The app (app.py) loads the fine-tuned DistilBERT from the models folder. If that folder is missing, it falls back to the pretrained RoBERTa model, so the app always runs. Each model is loaded once and kept in memory, and bulk results are cached, so changing a chart option does not re-run the models. It runs on a laptop CPU.

9.2 Single feedback analysis

The user pastes one review and clicks Analyze. The app shows:

- the overall sentiment and the model's confidence;
- a bar chart of how strongly the review relates to each business category;
- a table of aspect-level sentiment for the relevant categories.

The category list is editable in the sidebar, so the same app works for other businesses.

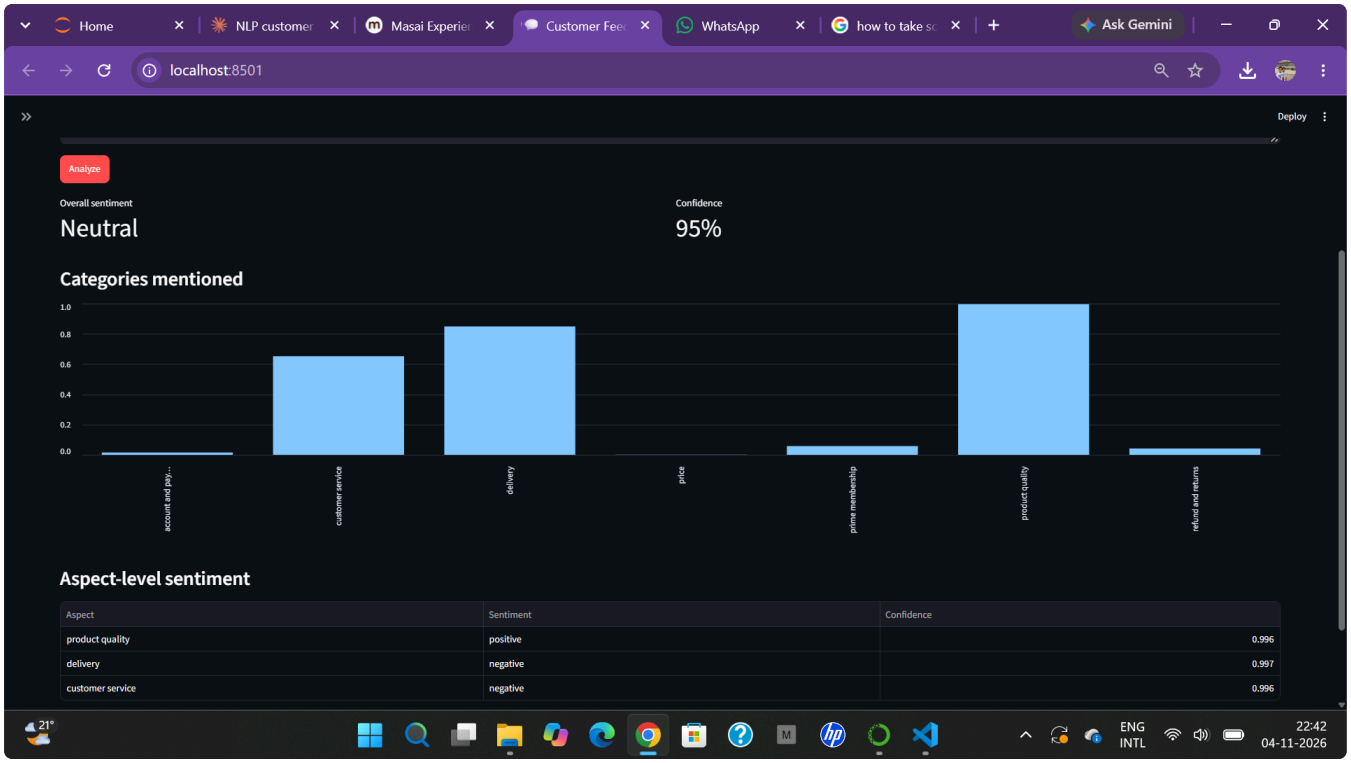


Figure 9.1: The Single feedback tab analyzing a mixed review.

9.3 Bulk analysis

The user uploads a CSV file. The app detects the review text and date columns (for Amazon_Reviews.csv, Review Text and Review Date) and analyzes a chosen number of rows. It shows:

- the number of reviews and the share of positive, neutral and negative ones;
- the sentiment distribution, and sentiment by category;
- a monthly or yearly sentiment trend;
- how often the predicted sentiment agrees with the customers' own star ratings;
- the ten most confidently negative reviews;
- a button to download all results as a CSV file.

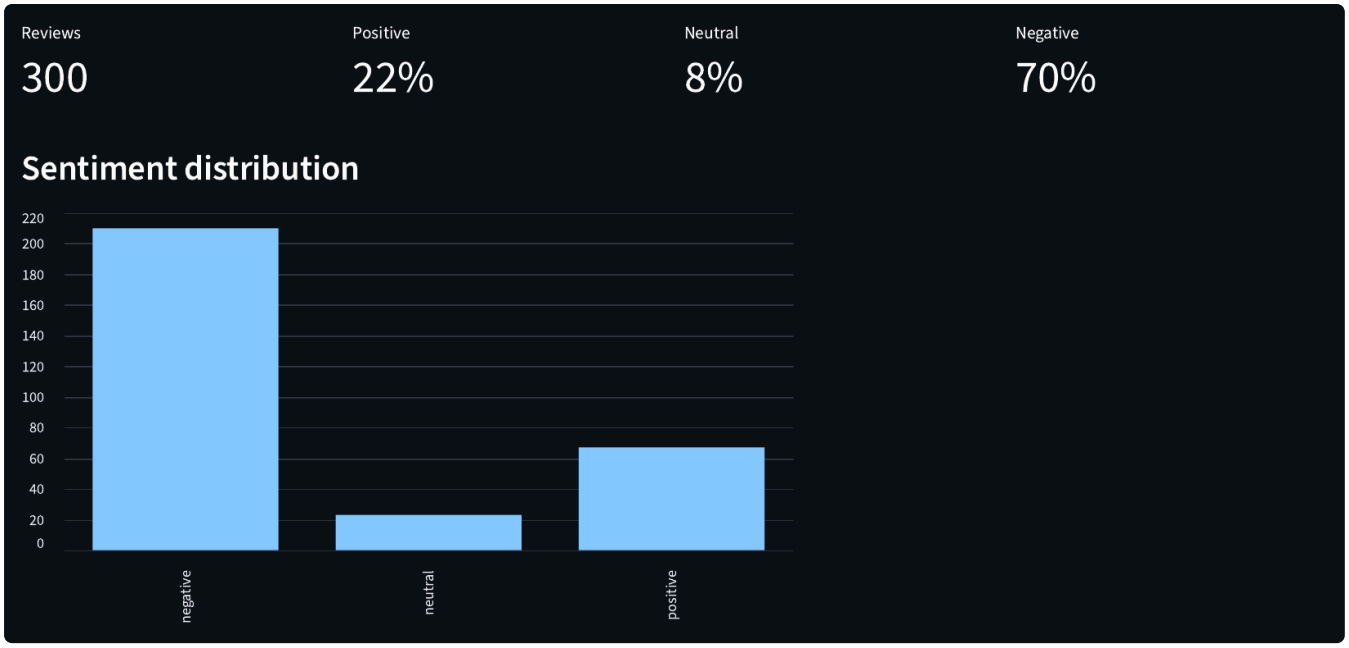


Figure 9.2: Bulk analysis of 300 randomly sampled reviews (sentiment only).

9.4 Bulk analysis results

The dashboard's predictions agreed with the customers' own star ratings for 93–98% of reviews in three separate runs, consistent with the model's 90.5% test accuracy.

Run	Reviews (random sample)	Category detection	Positive	Neutral	Negative	Agreement with star rating
1	300	Off	22%	8%	70%	93%
2	100	On	21%	5%	74%	96%
3	48	On	10%	6%	83%	98%

The largest run (300 reviews) predicted 70% negative reviews, close to the 68.6% share in the full dataset, so the dashboard gives a realistic picture even from a small sample. The smaller runs show higher agreement and a larger negative share, which is expected from random variation in small samples. Category detection runs the large BART-MNLI model once per review and category, which is slow on a laptop CPU, so it was run on samples of 100 reviews or fewer.

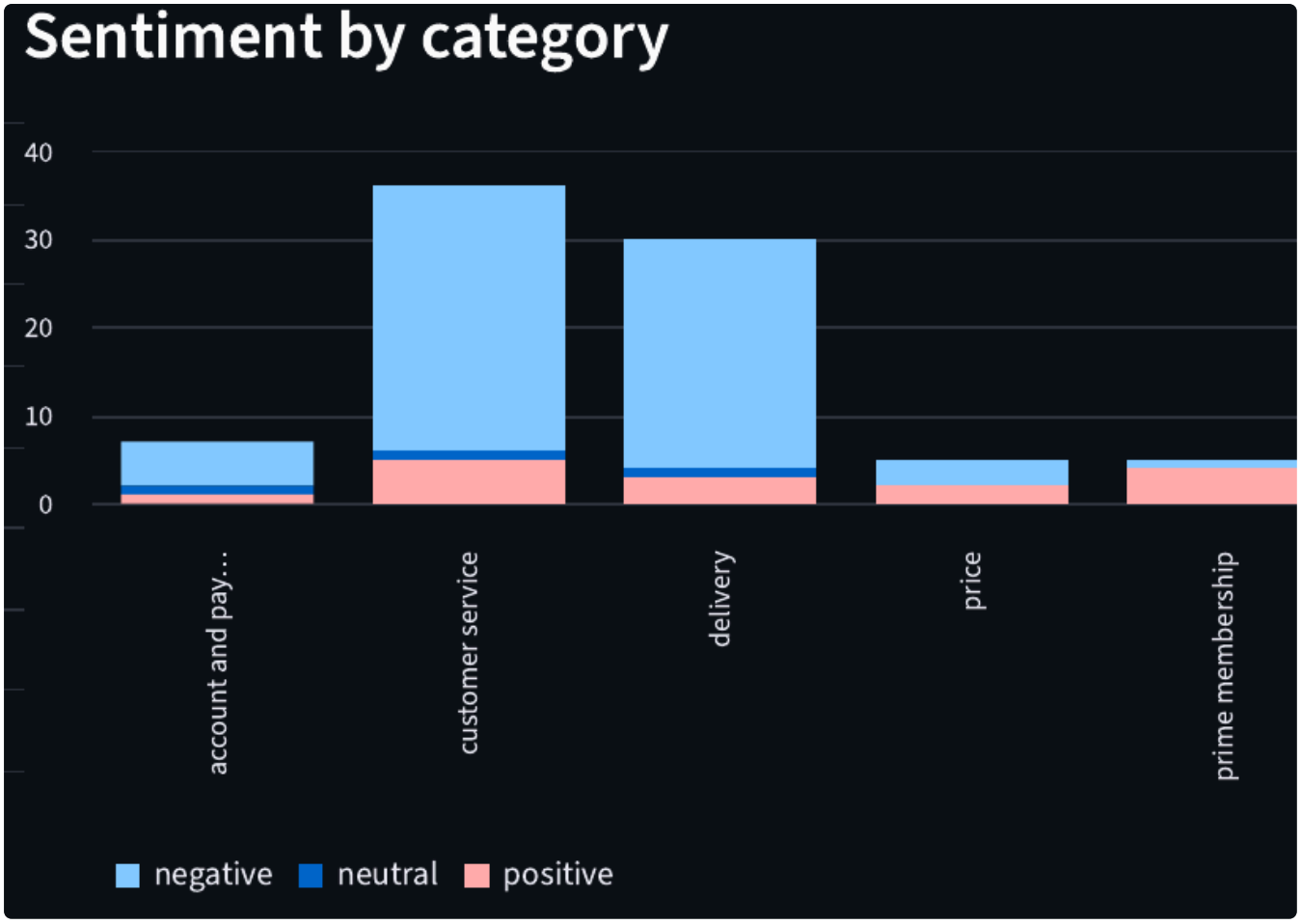
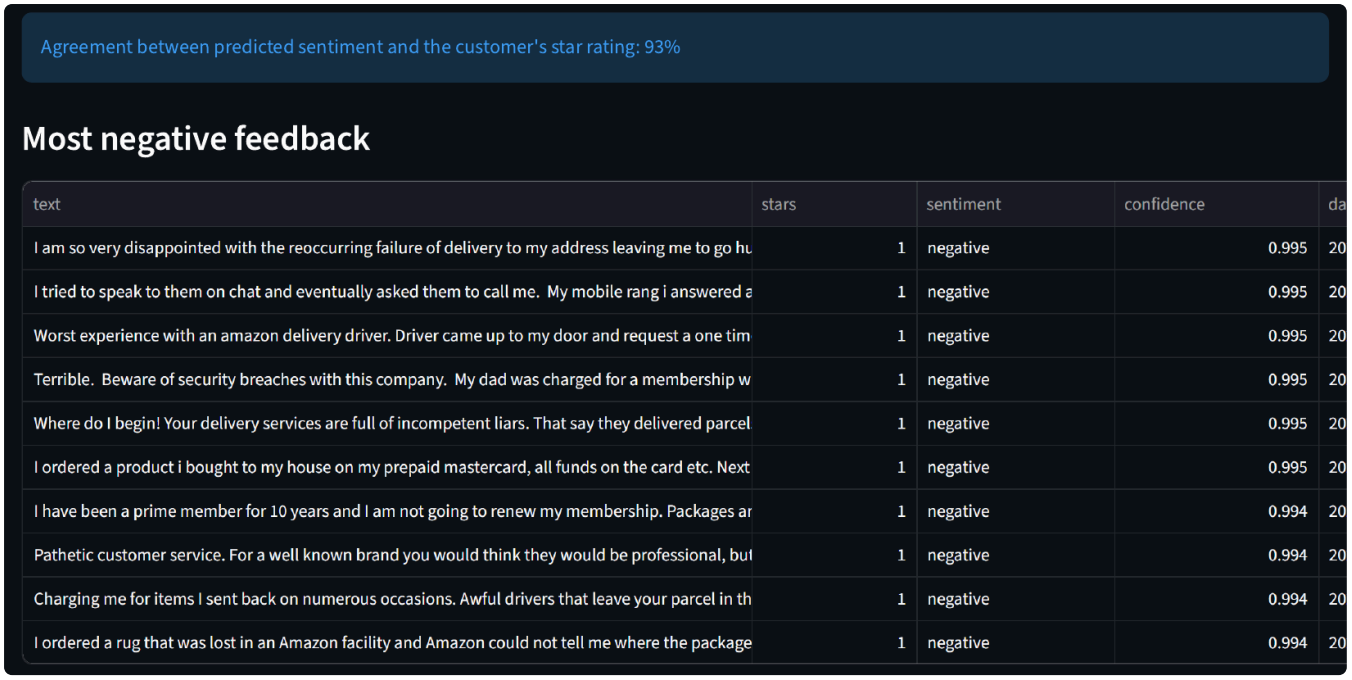


Figure 9.3: Sentiment by category for 100 randomly sampled reviews. Customer service and delivery are the most-mentioned categories, and most of their reviews are negative. The product quality and refund columns fall outside this screenshot.

Customer service (about 36 of 100 reviews) and delivery (about 30) dominate, matching the topic modeling results in Section 8.1, where support and delivery formed the largest groups of complaints.



Agreement between predicted sentiment and the customer's star rating: 93%

Most negative feedback

text	stars	sentiment	confidence	da
I am so very disappointed with the reoccurring failure of delivery to my address leaving me to go hu	1	negative	0.995	20
I tried to speak to them on chat and eventually asked them to call me. My mobile rang i answered a	1	negative	0.995	20
Worst experience with an amazon delivery driver. Driver came up to my door and request a one tim	1	negative	0.995	20
Terrible. Beware of security breaches with this company. My dad was charged for a membership w	1	negative	0.995	20
Where do I begin! Your delivery services are full of incompetent liars. That say they delivered parcel	1	negative	0.995	20
I ordered a product i bought to my house on my prepaid mastercard, all funds on the card etc. Next	1	negative	0.995	20
I have been a prime member for 10 years and I am not going to renew my membership. Packages ar	1	negative	0.994	20
Pathetic customer service. For a well known brand you would think they would be professional, but	1	negative	0.994	20
Charging me for items I sent back on numerous occasions. Awful drivers that leave your parcel in th	1	negative	0.994	20
I ordered a rug that was lost in an Amazon facility and Amazon could not tell me where the package	1	negative	0.994	20

Figure 9.4: Agreement with star ratings and the ten most confidently negative reviews (300-review run).

All ten most negative reviews have a 1-star rating and a confidence of 0.994–0.995. They describe failed deliveries, delivery drivers, unhelpful customer service, unexpected charges and lost parcels, the same problems identified by the topic model. A manager can read this table to find the most serious complaints without opening the full file.

10. Tools and Technologies

Everything in the project is free and open source, and it runs on free hardware.

Category	Tool	Used for
Language	Python 3.13	All code
Environment	Google Colab (free T4 GPU), Jupyter, VS Code	Training and development
Data	pandas, NumPy	Loading and processing data
Visualization	Matplotlib, seaborn, WordCloud	Charts and word clouds
Text processing	NLTK	Stopwords and lemmatization
Classical ML	scikit-learn	TF-IDF, Naive Bayes, Logistic Regression, Linear SVM, K-Means, metrics

Category	Tool	Used for
Lexicon	vaderSentiment	Rule-based baseline
Deep learning	PyTorch, Hugging Face Transformers, Datasets	Pretrained and fine-tuned transformers
Embeddings	Sentence-Transformers	all-MiniLM-L6-v2 embeddings
Topic modeling	BERTopic	Topic discovery
Web app	Streamlit	Interactive dashboard

10.1 Pretrained models (Hugging Face Hub)

Model	Task
distilbert-base-uncased	Base for fine-tuned sentiment classifier
cardiffnlp/twitter-roberta-base-sentiment-latest	Pretrained sentiment baseline
sentence-transformers/all-MiniLM-L6-v2	Sentence embeddings for topics
facebook/bart-large-mnli	Zero-shot categorization
yangheng/deberta-v3-base-absa-v1.1	Aspect-based sentiment
facebook/bart-large-cnn	Summarization

11. Conclusion and Future Scope

11.1 Conclusion

This project built a complete, free pipeline that turns raw customer reviews into sentiment scores, themes and readable insights. On 20,913 Amazon reviews, a fine-tuned DistilBERT reached a macro F1 of 0.717, outperforming a rule-based model, three classical machine learning models and a pretrained transformer. The comparison showed that fine-tuning on in-domain data gives the largest improvement, and that accuracy alone is misleading when classes are imbalanced. Most of the gain came from the rare neutral class, where DistilBERT found 62 of 174 neutral test reviews, more than three times as many as any other model.

The insight stage showed that customer satisfaction with Amazon fell sharply after 2018, and that complaints centre on customer service, delivery and refunds. Finding and fixing the placeholder reviews during topic modeling also showed the value of inspecting intermediate results rather than trusting a pipeline blindly. The Streamlit dashboard makes the system usable by people with no programming knowledge. On a random sample of 300 reviews, its predictions agreed with the customers' own star ratings 93% of the time.

11.2 Limitations

- **Noisy labels:** sentiment comes from star ratings, which do not always match the review text.
- **Small neutral class:** only 4.2% of reviews are neutral, which limits every model's performance on that class. Even the best model finds only 36% of neutral reviews, and it labels 157 negative reviews as neutral.
- **Truncation:** the fine-tuned model reads only the first 256 tokens of long reviews.
- **One company, one language:** all reviews are about Amazon and written in English.
- **Summaries:** the news-trained summarizer tends to copy sentences rather than rewrite them.
- **Compute:** the free GPU limited training to two epochs and a base-size model.

11.3 Future scope

- **Indian languages and Hinglish:** use models such as IndicBERT to analyze feedback written in Hindi or mixed Hindi–English.
- **Better handling of the neutral class:** collect more three-star reviews, or train with a class-weighted loss.
- **Sarcasm detection:** add a dedicated model to catch positive words used negatively.
- **Longer context:** use a long-document transformer so that long complaints are read in full.
- **Real-time analysis:** process new reviews as they arrive and alert the business when negative sentiment spikes.
- **Conversational insights:** connect the results to a free open-source large language model so a manager can ask questions such as "What were the main delivery complaints last month?".

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